

1 Resource Allocation

Overview

Menu	Production facility management → Resource planning → Resource allocation
Transaction code	resal
Function authorization	resal

Purpose

The *resource allocation* displays the following for the selected resources:

- primary resources (workplaces, machines).
- secondary resources, i.e. resources that are required if you assign an operation to a workplace.

the planned use of the resources in a Gantt chart and the point in time and duration of their assignment. This information is shown in the graphic resource allocation:

- operations pertaining to production orders
- operations pertaining to maintenance orders
- planned locking of resources

In addition, the application provides a tabular overview of tooling.

Integration

The displayed resource allocation results from detailed planning in the HYDRA Shop Floor Scheduling module or the ERP system.



You cannot make changes to the assignments in this application. To change assigned resources, you have to (re-)plan operations in the graphic planning module or in the ERP system.

Requirements

Operations must be planned on a workplace and resources must be assigned to the operations as production resources and tools. The resources must be created in the resource stock (WRM module).

Selection criteria

The application provides the following selection criteria:

Resource tab

Resource type (R)

Resource type of the resource. You can select multiple options.

Resource (R)

This selection criterion refers to the resource number. You can also use wildcards (placeholders *).

Assignment from / to (A)

Use the date selection to restrict the period of time for the data you want to evaluate.

You can also select past periods.

Designation (name) (R)

Name of the resource according to the resource configuration.

Resource status (R)

This selection criterion refers to the current statuses of the resource. You can select multiple options.

Active (R)

Use this tri-state checkbox to select:

- currently active resources (i.e. they are logged on to a workplace)
- currently inactive resources (i.e. they are not logged on to a workplace)
- currently active and inactive resources

Responsibility area (R)

Responsibility area of the resource according to the resource configuration.

Cost center (R)

Cost center of the resource according to the resource configuration.

Storage location (R)

"Original" storage location of the resource according to the resource configuration.

Current storage location (R)

Storage location the resource is currently assigned to.

Workplace from / to (A)

Workplace on which resources are planned via the production resource/tool list.

Short designation (name) (A)

Short name of the workplace according to the resource configuration.

Group from ... to (A)

Group of the workplace according to the resource configuration.

Order (A)

Order number of the operation that is planned on a workplace.

MES order number (A)

MES order number (combined order/OP number) of the operation that is planned on a workplace.

Resource family (R)

Resource family of the resource according to the configuration.

User field 1...6 (R)

Slow user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

User fields tab**Object type (R)**

Object type used to configure the user field key. The object type corresponds to the resource type.

Note: The selection list also shows object types that are not assigned to resources.

User field key (R)

User field key combining the direct user fields.

User fields (R)

Direct user fields that are assigned to the user field key according to the configuration.

The application identifies the assigned resources as follows:

Assignment of secondary resources based on planned operations

The application determines all assignments coinciding with the selected period.

1. The application identifies all planned operations with a planned start or planned end coinciding with the selection period. The operations must not be finished (control indicator <> E, D). The system also integrates the responsibility area of the workplace where the corresponding operation is planned.

The system always selects the operations logged on.

The system does neither integrate split masters nor individual OPs pertaining to merged operations generated via the MOC.

2. The selection includes the selection criteria identified with (A) above in the document.

3. For these (planned) operations, the system selects the resources that are assigned as production resources and tools. The selection includes the selection criteria identified with (R) above in the document. The selection does not include resources for which the user is not authorized via the responsibility area.

Assignment of primary resources (workplaces/machines) based on planned operations

The application determines all assignments coinciding with the selected period.

1. See description of secondary resources.
2. See description of secondary resources.
3. For these (planned) operations, the system selects the workplaces that are assigned to the operation (as workplace). The selection includes the selection criteria identified with (R) above in the document. The selection does not include resources for which the user is not authorized via the responsibility area.

Assignments based on locked resources

1. The application determines all locked resources (irrespective of whether it is a workplace/machine or other resource) according to the selection criteria identified with (R) in the table above.
2. The selection includes the locked resources with a start or end date of the lock coinciding with the selected period. The selection does not include resources for which the user is not authorized via the responsibility area.

Toolbar



Expand/collapse all groups

Click this button to expand or collapse all resource families (groups).



Zoom in

Click this button to increase the graphic use of resources (Gantt chart).



Zoom out

Click this button to reduce the graphic use of resources (Gantt chart).

Detail application *Resource Allocation*

The Gantt chart shows the resource allocation. You can use the *zoom in* and *zoom out* buttons to adjust the graphic presentation to the required size.



Changes to the Gantt chart are not saved.

Left-hand side

The resource family represents the grouping. Use the button  to collapse or the button  to expand the groups. The application shows all resources matching the entered selection criteria below the resource family.

The table shows the following information for the single resources:

- Current status information on the respective resource.
The application shows the configured color from the status configuration of resources for the current status.
- Resource number
- Current maintenance status (see the description below).
- Resource name

Current maintenance status

The table shows a symbol if a maintenance is due for a resource. The symbol color depends on the current status of the maintenance (depending on the maintenance limits that are exceeded). Currently, the following maintenance statuses are available:

- No maintenance due: the table does not show a symbol.
- Maintenance is due / level 1: blue symbol 
- Maintenance is due / level 2: yellow symbol 
- Maintenance is due / level 3: red symbol 

If several maintenance activities are due at the same time, the application shows the symbol of the maintenance with the highest priority. The priority sequence is as follows: no maintenance (lowest priority) → blue → yellow → red (highest priority).

For additional information on maintenances, please refer to the appropriate documentation.

Tooltip

If you move the mouse pointer over a resource, a tooltip providing the following information appears:

- Resource type
- Resource
- Name (of the resource).
- Current status (number of the resource status, status text)
- Status since (date, time)
- Current storage location
- Active: yes/no

Context menu

You can start the following applications via the context menu of the resource:

- Resource overview (resov)
- Activity schedule (rmcal)
- Workplaces/machines (wpov);
only if resource type = MNR

Right-hand side

In the resource lines, the chart shows the planned operations (production orders), the planned maintenance operations and the locked resources as horizontal bars. The resource family lines are shown in gray and do not contain any graphic bars.

The bar length results from the "planned start" and "planned end" period of operations (maintenance and production operations). The bar length of locked resources results from the start and end time of resource locks.

Depending on the planning, one line may include several operation bars, if tools have been planned several times simultaneously. In this case, the bars do not overlap but are displayed one underneath the other.

Planned operations with a planned start in the past

The system integrates assignments for planned operations whose planned start is in the past as follows:

- The system determines the delta between the planned end and the planned start (synchronized with the Gregorian calendar).

- The system sets the planned start of the assignment to "now".
- The system sets the planned end to the planned start ("now") + the delta.
- The operation bar is shown hatched.
- The tooltip shows "Start date of OP changed".



In contrast to the graphic planning board (HLS), the bar lengths do not result from the planned start + remaining run time, but exclusively from the planned dates.

Operations logged on

The graphic display also shows allocations of operations that are currently logged on. For this purpose, the application shows the bars from the start point (of the selection period) to the planned end of the operation.



In contrast to the graphic planning board (HLS), the bar lengths do not result from the planned start + remaining run time, but exclusively from the planned dates. A certain uncertainty occurs, if an operation is logged on before its planned start or if the actual remaining run time differs from the planned remaining run time.

Bar colors

These are the bar colors used:

- Locked resources: red
- Maintenance operations (order type category: PM): light blue
- All other operations: the bar color results from the current operation status and the color defined in status assignment.

Bar legend

- Operation bars: workplace number / group
- Locked resources: no bar text

Tooltip of the operation bar

When you move your mouse cursor over an operation bar, a tool tip providing the following information appears:

- Workplace

- Group
- MES order number (of the operation)
- Article (of the operation)
- Operation status (status text)
- Resource (number, name)
- Demand (according to the production resources and tools list)
- Assignment from (date, time): planned start according to order backlog
- Assignment to (date, time): planned end according to order backlog
- "Start date of OP changed" (see the description for operations with a planned start in the past).

Tooltip of locked resources

If you move the mouse pointer over a locked resource, a tooltip providing the following information appears:

- Resource
- Designation (name)
- Lock from (date, time)
- Lock till (date, time)

Context menu of operation bars

You can start the following applications via the context menu of operation bars:

- Order information (orin)
- Workplaces/machines (wpov)
- Resource overview (resov)

Context menu of locked resources

If a resource is locked, you can only open the resource overview (resov).

Detail application *Resource staging list*

Since the main application area of this detail application is the presentation of the provision of tools, this detail application is called "resource staging list. Nonetheless, this application can also be used for other resources ("resource staging list"). The list shows the planned usage dates of the resources in tabular form. The detail application is docked "behind" the graphic display.

In contrast to the graphic presentation, the list does not show any locks.

The detail application provides the following information:

Category	Field	Description
Resource	Resource type	Type of resource
	Resource	Technical, unique name of the resource
	Resource family	Family assigned to the resource
	Responsibility area	Responsibility area of the resource
	Short name	Short name of the resource, the field is only filled for resources of the type MNR.
	Designation (name)	Name of resource
	Status	Status number of the current status of the resource.
	Status	Status text of the current resource status - highlighted in color
	Active	Identifies whether a resource is active
	Cost center	Cost center of the resource.
Assignment	Assignment from	Date + time
	Assignment to	Date + time
	Assignment from date	Date (only)
	Assignment to date	Date (only)
Workplace	Workplace	Workplace where the operation the resource is assigned to is planned.
	Short name	Short name of the workplace.
	Designation (name)	Name/description of the workplace.
	Cost center	Cost center of the workplace
Order	Order type	Order type of the operation.
	Order	Order number of the operation.
	Sequence	Sequence number of the operation, if active.
	OP	Operation number of the operation.
	Split	Split number of the operation, if the operation is split.
	SOP	Sub-operation number (reserved)
	MES order number	Combined order/operation number.
	Article	Articles of the operation
	Article name/designation	Article name of the operation.
	OP name/designation	Operation name.